

Assignment 5 — Write-up

DSC 190/291 — Learning Theory Student: Zeyu Bian

Part A — Sparse Linear Predictors: Statistical Sparsity vs Agnostic Hardness

Throughout, $\mathcal{H}_{d,k} = \{x \mapsto \text{sign}(\langle w, x \rangle) : w \in \mathbb{R}^d, \|w\|_0 \leq k\}$, with $\text{sign}(0) = +1$.

A.1 What Homework 4 already gives

Statistical side. Homework 4 showed $\text{VCdim}(\mathcal{H}_{d,k}) = O(k \log(ed/k))$ for $k \leq d/2$ (and $O(d)$ for larger k). The standard agnostic VC sample-complexity bound (e.g. SSBD §6.4) then gives that if *exact* ERM over $\mathcal{H}_{d,k}$ were available, then for every $\mathcal{D}$, with probability $\geq 1 - \delta$,

$$L_{\mathcal{D}}(\hat{h}_{\text{ERM}}) \leq \inf_{h \in \mathcal{H}_{d,k}} L_{\mathcal{D}}(h) + C \sqrt{\frac{k \log(ed/k) + \log(1/\delta)}{n}},$$

so $n = O((k \log(ed/k) + \log(1/\delta))/\varepsilon^2)$ samples suffice to learn $\mathcal{H}_{d,k}$ to excess risk ε . **The information is in the data.**

Computational side. The brute-force realizable algorithm enumerates supports $T \subseteq [d]$ with $|T| \leq k$ ($\binom{d}{0} + \dots + \binom{d}{k} \leq (ed/k)^k$ supports) and solves a halfspace feasibility LP on each, in $\text{poly}(k, n)$ time. Total:

$$\binom{d}{\leq k} \cdot \text{poly}(k, n) = O((ed/k)^k \cdot \text{poly}(k, n)).$$

Polynomial in d iff $k = O(1)$. For $k = \Theta(\log d)$ this is $d^{\Theta(\log d)}$, superpolynomial.

No contradiction. Sample complexity and runtime are different resources. Knowing how many examples it would take to identify the best h does not produce an algorithm that finds it. Part A.2 makes this concrete: under standard assumptions, no efficient algorithm exists.

A.2 Hardness of AGREEMENT via SETCOVER

SETCOVER (reminder). Input: universe $U = \{u_1, \dots, u_r\}$, sets $A_1, \dots, A_q \subseteq U$, integer $k \leq q$. Output: yes iff there exists $R \subseteq [q]$ with $|R| \leq k$ and $\bigcup_{j \in R} A_j = U$.

Reduction

Let $M = q + 1$. Build the AGREEMENT $_{\mathcal{H}_{d,k}}$ instance with $d = q$ and the same k :

- **Cost examples (q in total).** For each $j \in [q]$, one positive example $(e_j, +1)$, where $e_j \in \mathbb{R}^q$ is the j -th standard basis vector.
- **Coverage examples (Mr in total).** For each $u_i \in U$, M identical copies of the negative example $(a^{(i)}, -1)$, where $a^{(i)} \in \{0, 1\}^q$ has $a_j^{(i)} = \mathbf{1}[u_i \in A_j]$.

Total sample size $m = q + Mr$. Set the agreement threshold

$$K = (q - k) + Mr.$$

Mistake accounting for any classifier h_w with $\|w\|_0 \leq k$

Let $R(w) = \{j : w_j \neq 0\}$, $|R(w)| \leq k$.

- **Cost examples.** $(e_j, +1)$ is misclassified iff $\text{sign}(w_j) = -1$ iff $w_j < 0$. So the number of cost-example mistakes is $|\{j : w_j < 0\}| \leq |R(w)| \leq k$.
- **Coverage examples.** $(a^{(i)}, -1)$ is misclassified iff $\langle w, a^{(i)} \rangle = \sum_{j: u_i \in A_j} w_j \geq 0$. There are two ways this can happen:
 - u_i is **not** covered by $R(w)$ — i.e. $u_i \notin A_j$ for every $j \in R(w)$ — in which case the sum is $0 \geq 0$ and the mistake fires. Each uncovered u_i then contributes M mistakes (one per copy).
 - u_i is covered but the nonzero w_j 's on covering $j \in R(w)$ happen to sum to ≥ 0 . Still costs M mistakes.

Forward direction (SETCOVER YES $\Rightarrow$ AGREEMENT $\geq K$)

Suppose $R \subseteq [q]$ with $|R| \leq k$ and $\bigcup_{j \in R} A_j = U$. Define w by $w_j = -1$ if $j \in R$ and $w_j = 0$ otherwise. Then $\|w\|_0 \leq k$.

- Cost mistakes: $|R| \leq k$.
- Coverage mistakes: each u_i has at least one $j \in R$ with $u_i \in A_j$, so $\langle w, a^{(i)} \rangle = \sum_{j \in R: u_i \in A_j} (-1) = -|\{j \in R : u_i \in A_j\}| < 0$, classified correctly. **Zero** coverage mistakes.

Total mistakes $\leq k$, so agreement $\geq m - k = q + Mr - k \geq q - k + Mr = K$.

Reverse direction (AGREEMENT $\geq K \Rightarrow$ SETCOVER YES)

Suppose w achieves $\geq K$ agreements, so $\leq m - K = k$ mistakes. Define $R(w) = \{j : w_j \neq 0\}$, $|R(w)| \leq k$.

Suppose for contradiction that $R(w)$ does **not** cover U . Then some u_i is uncovered, contributing $M = q + 1$ coverage mistakes. But total mistakes $\leq k \leq q < M$, contradiction. So $R(w)$ covers U , and $|R(w)| \leq k$ — a valid set cover.

Conclusion: hardness of agnostic proper learning

SETCOVER is NP-hard, and the reduction above runs in time polynomial in r, q, k ($M = q + 1$ is polynomial; the constructed sample has $q + Mr = O(qr)$ examples in $\mathbb{R}^q$). So **AGREEMENT** $_{\mathcal{H}_{d,k}}$ is **NP-hard** when d and k are part of the input.

Week 5 learner-to-agreement theorem (paraphrased). If a class $\mathcal{H}$ has an efficient proper agnostic PAC learner, then the AGREEMENT problem for $\mathcal{H}$ is in randomized polynomial time (RP). Idea: feed the learner the empirical distribution $\hat{D}_S$ as the unknown distribution, with ε smaller than $1/m$, and its output achieves empirical 0-1 risk within ε of the optimum — close enough to decide AGREEMENT by checking the output on S .

Applying this to $\mathcal{H}_{d,k}$ with d, k both part of the input: an efficient proper agnostic PAC learner would put **AGREEMENT** $_{\mathcal{H}_{d,k}} \in \text{RP}$. Combined with NP-hardness of **AGREEMENT** $_{\mathcal{H}_{d,k}}$, this would give $\text{NP} \subseteq \text{RP}$. **Conclusion:** unless $\text{NP} = \text{RP}$, no efficient proper agnostic PAC learner exists for sparse halfspaces when k is part of the input.

A.3 Experiment and interpretation

The companion script `experiment.py` runs:

1. **Proper sparse 0-1 baseline.** For each support $T \subseteq [d]$ with $|T| \leq k$, fit a halfspace by direct ERM on the $|T|$ -dimensional subproblem. We use Perceptron in the realizable region and an ℓ_2 -LR proxy on the support in the noisy region (proper, then evaluated under 0-1 loss). The baseline is **approximate** — within each support we minimize a convex surrogate, but the *enumeration over supports* is exact. Across supports we report the best 0-1 agreement found.
2. **Convex surrogate.** ℓ_1 -regularized logistic regression (`sklearn.linear_model.LogisticRegression(penal solver='liblinear')`), sweeping the regularization strength C and reporting the best 0-1 agreement over the sweep. This does **not** produce a k -sparse classifier in general — the ℓ_1 penalty only shrinks coefficients; the support size depends on C and the data, and may be larger or smaller than k .
3. **Runtime scaling** in d (with $k = 3$ fixed) and in k (with $d = 20$ fixed).
4. **Agreement comparison** on a synthetic instance generated by planting a sparse target plus label noise.

The figures below summarize the results. Concrete numbers from a representative run ($n = 200$ for the runtime sweeps, $n = 400$ for the noise sweep):

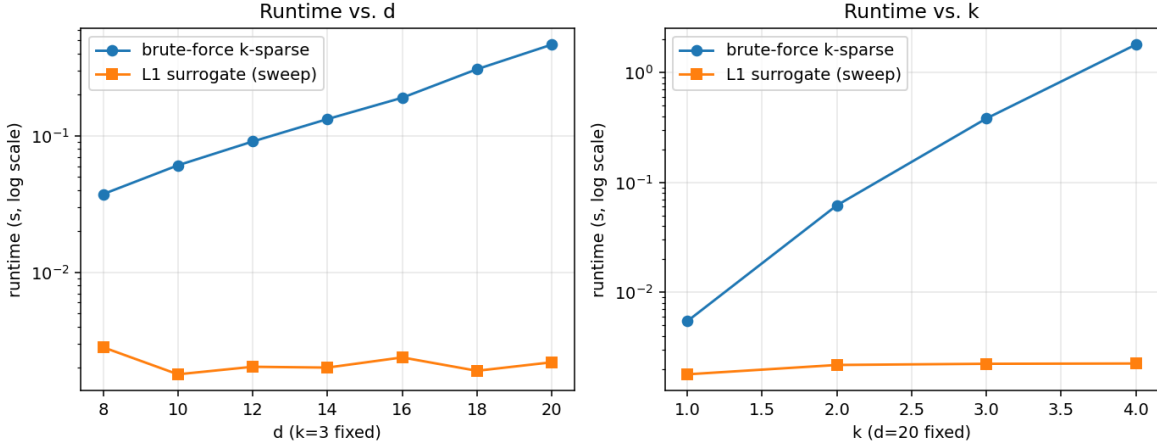


Figure 1: Runtime scaling: brute-force k -sparse enumeration vs. L_1 -regularized logistic regression. Left: varying d with $k=3$ fixed. Right: varying k with $d=20$ fixed. Log y-axis.

- **Runtime in d (with $k = 3$).** Brute force: $0.04s \rightarrow 0.42s$ as d goes $8 \rightarrow 20$. ℓ_1 surrogate: essentially constant at $\sim 0ms$ resolution.
- **Runtime in k (with $d = 20$).** Brute force: $0.01s (k=1) \rightarrow 0.06s (k=2) \rightarrow 0.39s (k=3) \rightarrow 1.81s (k=4)$ — roughly $5\times$ per unit of k , matching the $\binom{20}{k}$ growth. Surrogate flat.
- **Agreement vs. noise.** Both methods are essentially indistinguishable on this synthetic generator across noise rates $0 \rightarrow 0.4$, sometimes one wins by a handful of examples out of 400, sometimes the other (e.g. at noise = 0.3: brute 267/400 vs. ℓ_1 275/400; at noise = 0.4: 241 vs 237).

Why agreement is close even though the problems differ. Our brute-force baseline itself uses an ℓ_2 -regularized logistic regression as the per-support solver, so the *true* 0-1 agreement of the best k -sparse halfspace is upper-bounded by what the baseline reports — the baseline already loses

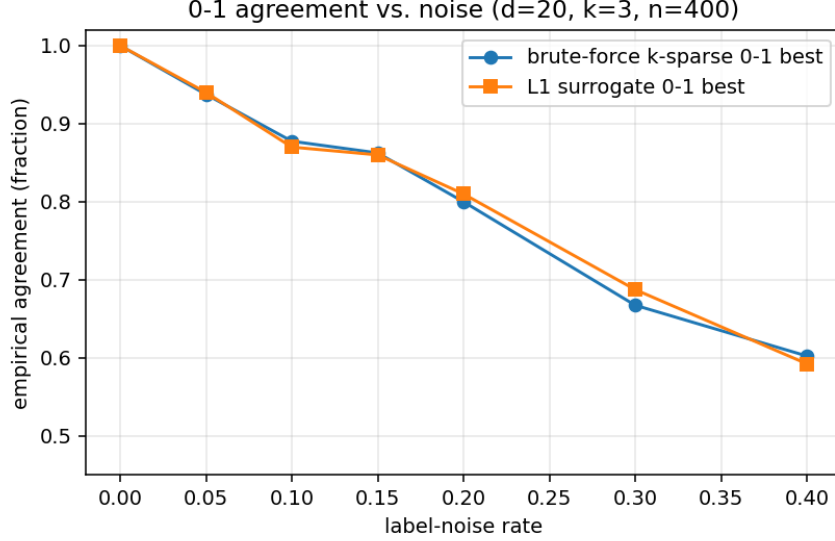


Figure 2: Empirical 0-1 agreement as a function of label-noise rate, $d=20$, $k=3$, $n=400$.

some 0-1 mass to a convex surrogate *inside* each support. A truly exact 0-1 baseline would require enumerating dichotomies on each support (exponential in n), which we avoid here.

Runtime story. Even with this approximation, the runtime gap is exactly the predicted $\binom{d}{k}$ blow-up: brute-force is already $\sim 50\times$ slower than the surrogate at $d = 20, k = 4$, and the constant scales with the number of supports. The ℓ_1 surrogate runs in $\text{poly}(d, n)$ regardless.

What the experiment illustrates. Empirical 0-1 agreement over $\mathcal{H}_{d,k}$ and convex ℓ_1 -surrogate optimization are *different objectives*. The first is what the agreement decision problem asks for and is NP-hard (A.2); the second is convex, efficient, and statistically clean, but (i) does not enforce hard sparsity and (ii) optimizes a Lipschitz surrogate that need not align with 0-1 risk (Part B.2). On this benign synthetic distribution (Gaussian features, planted sparse target, symmetric label noise) the surrogate happens to track the brute-force baseline closely in 0-1 agreement; that’s a feature of the *distribution*, not a guarantee in general. The surrogate method does **not** return a proper k -sparse classifier — inspecting the output of the ℓ_1 sweep typically yields a vector with strictly more than k nonzeros, and there is no guarantee it returns the 0-1-best one within the sparsity budget. The Part B.2 counterexample shows that there exist distributions where the surrogate’s 0-1 agreement is *much* worse than the best sparse halfspace; the experiment here is consistent with that statement, not a refutation of it.

Part B — Convex Fixed-Feature Learning and Its Limits

B.1 Hinge-loss + ℓ_1 as a linear program

Variables. $w_j^+, w_j^- \geq 0$ for $j = 1, \dots, d$; $\xi_i \geq 0$ for $i = 1, \dots, m$.

LP.

$$\begin{aligned}
& \text{minimize} && \frac{1}{m} \sum_{i=1}^m \xi_i \\
& \text{s.t.} && \xi_i \geq 1 - y_i \sum_{j=1}^d (w_j^+ - w_j^-) x_{i,j} \quad (i = 1, \dots, m) \\
& && \xi_i \geq 0 \quad (i = 1, \dots, m) \\
& && \sum_{j=1}^d (w_j^+ + w_j^-) \leq B \\
& && w_j^+, w_j^- \geq 0 \quad (j = 1, \dots, d)
\end{aligned}$$

Role of each family.

- $w_j = w_j^+ - w_j^-$ with $w_j^\pm \geq 0$ is the standard split into positive/negative parts. At any optimum, $\min(w_j^+, w_j^-) = 0$ (otherwise reduce both by $\min$ without changing w_j , decreasing $w_j^+ + w_j^-$ slack on the ℓ_1 budget), so $|w_j| = w_j^+ + w_j^-$.
- The two ξ_i inequalities linearize the hinge: at optimum $\xi_i = \max(0, 1 - y_i \langle w, x_i \rangle)$, since the objective pushes each ξ_i down to the binding lower bound.
- The $\sum (w_j^+ + w_j^-) \leq B$ constraint linearizes the ℓ_1 ball $\|w\|_1 \leq B$ via the split.

The LP has $O(d+m)$ variables and $O(d+m)$ constraints; standard interior-point or simplex methods solve it in $\text{poly}(d, m)$ time.

Why this is not solving 0-1 sparse agreement. Two reasons.

1. **Loss.** Hinge is a (convex) upper bound on $\mathbf{1}[yf \leq 0]$, not a faithful surrogate. Part B.2 below exhibits a distribution where every hinge minimizer is a 0-1 disaster.
2. **Constraint.** $\|w\|_1 \leq B$ is the *convex relaxation* of $\|w\|_0 \leq k$, not the same thing. ℓ_1 encourages sparsity but does not enforce a hard support-size bound; nothing in the LP forces the output to have $\leq k$ nonzeros.

So even at its optimum, the LP solves a convex surrogate problem that is neither in $\mathcal{H}_{d,k}$ nor optimizing 0-1 loss.

B.2 A concrete counterexample

Fix $p \in (0, 1/2)$ and $M > (1-p)/p$. $\mathcal{D}_{p,M}$ puts mass $1-p$ on $(1, +1)$ and mass p on $(-M, +1)$. Predictors $f_w(x) = wx$. By Part B's convention, $\ell_{0-1}(f_w(x), y) = \mathbf{1}[yf_w(x) \leq 0]$ (zero margin counts as error).

Population hinge risk

$$L^{\text{hinge}}(w) = (1-p)(1-w)_+ + p(1+Mw)_+.$$

The breakpoints of the two hinges are at $w = 1$ and $w = -1/M$. Compute on three regions:

- $w \leq -1/M$: $1-w \geq 1+1/M > 0$ and $1+Mw \leq 0$, so $L = (1-p)(1-w)$, slope $-(1-p) < 0$, *decreasing* in w .
- $-1/M \leq w \leq 1$: both hinges active, $L = (1-p)(1-w) + p(1+Mw) = 1 + (pM - (1-p))w$, slope $pM - (1-p) > 0$ since $M > (1-p)/p$. *Increasing* in w .
- $w \geq 1$: $L = p(1+Mw)$, slope $pM > 0$, *increasing* in w .

So L^{hinge} decreases then increases, with a unique global minimum at

$$w^* = -\frac{1}{M}, \quad L^{\text{hinge}}(w^*) = (1-p)(1+1/M).$$

0-1 risks

For any w :

$$L_{\mathcal{D}}^{0-1}(f_w) = (1-p)\mathbf{1}[w \leq 0] + p\mathbf{1}[-Mw \leq 0] = (1-p)\mathbf{1}[w \leq 0] + p\mathbf{1}[w \geq 0].$$

- $w > 0$: $L^{0-1} = p$.
- $w < 0$: $L^{0-1} = 1 - p$.
- $w = 0$: both indicators fire, $L^{0-1} = 1$.

Hence

$$\inf_w L_{\mathcal{D}}^{0-1}(f_w) = p,$$

attained at any $w > 0$.

The unique hinge minimizer is $w^* = -1/M < 0$, so

$$L_{\mathcal{D}}^{0-1}(f_{w^*}) = 1 - p.$$

Choice of p, M for the gap

Given $\varepsilon > 0$ and $\alpha < 1$: pick p with $0 < p \leq \varepsilon$ and $p \leq 1 - \alpha$ (both possible since $\varepsilon, 1 - \alpha > 0$; take $p < \min(\varepsilon, 1 - \alpha, 1/2)$), then any $M > (1 - p)/p$. We have

$$\inf_w L_{\mathcal{D}}^{0-1}(f_w) = p \leq \varepsilon, \quad L_{\mathcal{D}}^{0-1}(f_{w^*}) = 1 - p > \alpha.$$

Moral. A *single* heavy-margin negative-side example with mass p can drag the hinge minimizer to the wrong sign even though that sign is wrong on $1 - p$ of the mass. Hinge minimization is sensitive to large-magnitude features in a way 0-1 risk is not.

B.3 The fixed-feature parity barrier

Lower bound $D \geq 2^d$

Form the $2^d \times 2^d$ matrix H with $H_{I,x} = \chi_I(x) = \prod_{i \in I} x_i$, rows indexed by $I \subseteq [d]$, columns by $x \in \{-1, +1\}^d$.

Orthogonality of rows. For $I, J \subseteq [d]$,

$$(HH^\top)_{I,J} = \sum_x \chi_I(x) \chi_J(x) = \sum_x \chi_{I \Delta J}(x),$$

since $\chi_I \chi_J = \chi_{I \Delta J}$ on $\{\pm 1\}^d$. For $K = I \Delta J \neq \emptyset$ the sum is $\sum_{x \in \{\pm 1\}^d} \prod_{i \in K} x_i = \prod_{i \in K} (\sum_{x_i \in \{\pm 1\}} x_i) = 0$. For $K = \emptyset$ the sum is 2^d . So $HH^\top = 2^d \cdot I_{2^d}$ and $\text{rank}(H) = 2^d$.

Factorization. The representation assumption $\chi_I(x) = \langle w_I, \varphi(x) \rangle$ gives the matrix factorization $H = W\Phi$, where $W \in \mathbb{R}^{2^d \times D}$ has rows $w_I^\top$ and $\Phi \in \mathbb{R}^{D \times 2^d}$ has columns $\varphi(x)$.

Therefore

$$2^d = \text{rank}(H) = \text{rank}(W\Phi) \leq \min(\text{rank}(W), \text{rank}(\Phi)) \leq D.$$

Hence $\boxed{D \geq 2^d}$.

Tight upper bound: $D = 2^d$ suffices

Define $\varphi : \{-1, +1\}^d \rightarrow \mathbb{R}^{2^d}$ by $\varphi(x) = (\chi_I(x))_{I \subseteq [d]}$. For any $J \subseteq [d]$, take $w_J = e_J \in \mathbb{R}^{2^d}$ (the indicator of coordinate J). Then

$$\langle w_J, \varphi(x) \rangle = \sum_I (e_J)_I \chi_I(x) = \chi_J(x).$$

So every parity is realized exactly by a linear predictor in this 2^d -dimensional feature space, matching the lower bound.

Week 5 message

Fixed-feature convex learning has a clean separation: *given* a feature map, the empirical risk is convex (hinge, logistic, squared loss) and we can solve it efficiently with VC/Rademacher-style statistical guarantees. But the *representational power* is capped by the feature dimension D . For families like parities, D must be exponential in d — which sinks both sample complexity ($\text{VCdim} \leq D$ but we need $D = 2^d$) and computation (the feature vector has 2^d entries). This is why fixed features hit a wall on simple-looking combinatorial concept classes, and motivates *learning* the features (Part B.4 / neural nets) — at the cost of convexity.

B.4 Same predictors, different optimization geometry

Same hypothesis class

$f_{u,v}(x) = v\langle u, x \rangle = \langle vu, x \rangle$, so every (u, v) realizes the linear predictor $\beta = vu$. Conversely, for any $\beta \in \mathbb{R}^d$ take $u = \beta, v = 1$; then $vu = \beta$. So

$$\{f_{u,v} : u \in \mathbb{R}^d, v \in \mathbb{R}\} = \{f_\beta : \beta \in \mathbb{R}^d\}.$$

L_{lin} is convex

$L_{\text{lin}}(\beta) = \frac{1}{m} \sum_i (\langle \beta, x_i \rangle - y_i)^2 = \frac{1}{m} \|X\beta - y\|^2$, a quadratic with Hessian $\frac{2}{m} X^\top X \succeq 0$. Convex.

L_{net} is not convex (Jensen violation)

Take $d = 1, m = 1, x_1 = 1, y_1 = 1$. Then $L_{\text{net}}(u, v) = (vu - 1)^2$. Evaluate at three points: - $(u_1, v_1) = (1, 1)$: $L_{\text{net}} = 0$. - $(u_2, v_2) = (-1, -1)$: $L_{\text{net}} = 0$. - Midpoint $(\bar{u}, \bar{v}) = (0, 0)$: $L_{\text{net}} = (0 - 1)^2 = 1$.

If L_{net} were convex, $L_{\text{net}}(\bar{u}, \bar{v}) \leq \frac{1}{2} L_{\text{net}}(u_1, v_1) + \frac{1}{2} L_{\text{net}}(u_2, v_2) = 0$. But $L_{\text{net}}(0, 0) = 1 > 0$. **Jensen violated**, so L_{net} is not convex.

Global minima coincide

The set of predictors is the same, so $\inf_{u,v} L_{\text{net}}(u, v) = \inf_\beta L_{\text{lin}}(\beta)$. If β^* minimizes L_{lin} and $\beta^* = vu$, then $L_{\text{net}}(u, v) = L_{\text{lin}}(vu) = L_{\text{lin}}(\beta^*) = \inf L_{\text{lin}} = \inf L_{\text{net}}$, so (u, v) is a global minimizer of L_{net} .

What this shows about learned features

The two parameterizations realize the *same* hypothesis class with the *same* infimum of empirical risk and the *same* set of global minima (up to factorization). But the *optimization landscapes are*

different: L_{lin} is convex, L_{net} is not. Reparameterizing the linear function $x \mapsto \langle \beta, x \rangle$ as a product $x \mapsto v \langle u, x \rangle$ already breaks convexity, even though we have gained no expressive power.

This is a miniature version of the deep-learning story: turning features from *fixed* (where the empirical risk is convex in the predictor) into *learned* (where the network parameterization composes linearities into non-convex products and adds nonlinearities) destroys convex tools. The hypothesis class can be the same, or larger, but the algorithmic problem is fundamentally different — gradient methods on non-convex losses have no a-priori convergence guarantee to the global optimum, even when the global optimum is known to exist and coincide with a convex problem's optimum.